

The Blueprint & The Toy: Intro to Classes & Objects



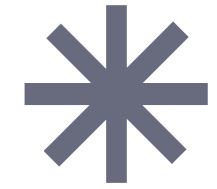


The Hook: LEGO vs. Instruction Manual



Imagine if you had LEGO blocks but no instructions... How do you know what to build? A Class is like the instruction manual — it tells you how to build something!





What is a Class? The Blueprint

A Class is a blueprint — like a cookie cutter for making Sweet dumpling !

- Blueprint means a plan or shape
- Helps make many things that look the same



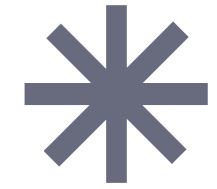


What is an Object? The Actual Item

An Object is the actual thing made from the blueprint — like a Maruti Suzuki Swift car parked in your society!

- Objects are created using the blueprint
- Each object is a real thing you can see and use





Gujiya Cookie Cutter & Cookies

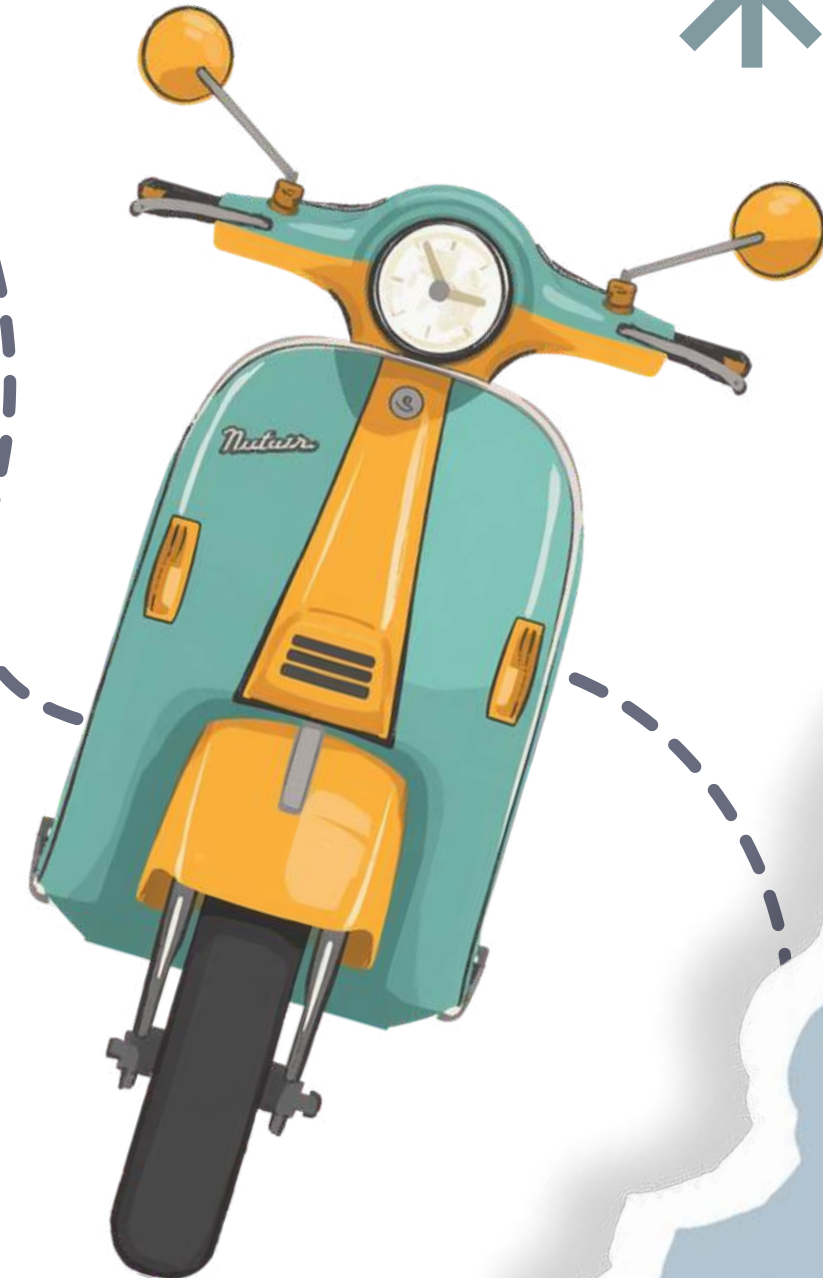
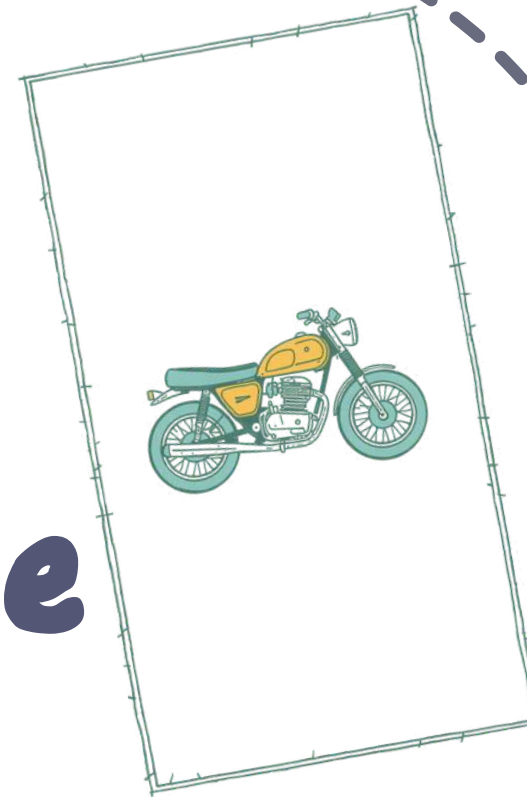


- Class = Sweet dumpling cookie cutter (blueprint)
- Object = Sweet dumplingcookies made using the cutter
 - Attribute example: Shape, Size (small, big)
- Method example: Bake() - makes the cookie ready!



Hero Motocorp Bike Factory Example

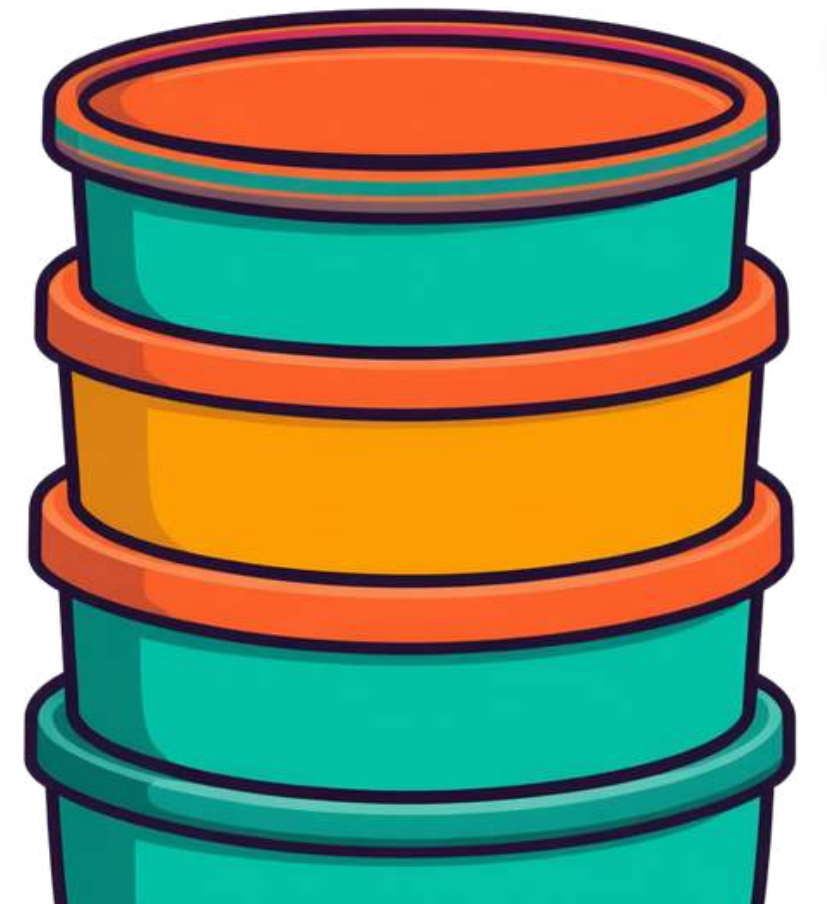
Imagine a Hero bike factory. The bike design
is the Class. Each bike made is an Object.
Class = Bike design | Object = The bike you
ride





Attributes & Methods

Attributes are what an object HAS.
Methods are what an object DOES.
Let's explore with everyday Indian examples!



Hands-On Activity: Let's Code a Student Class!

Let's create a Class called Student! This is our blueprint that tells Python how to make student objects.





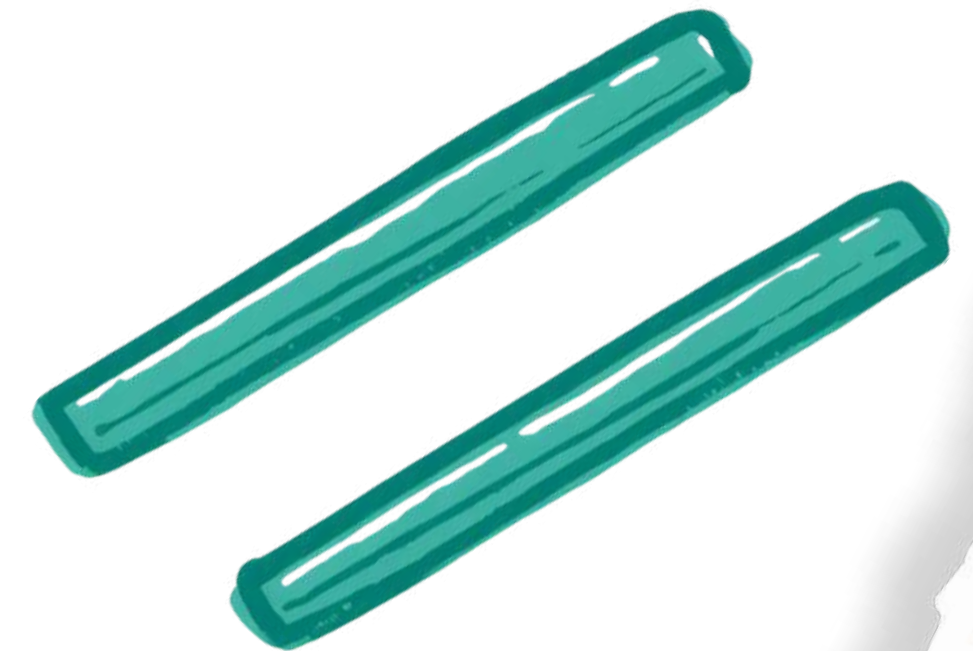
Creating an Object from the Student Class

Now, let's create an actual student named Kabir!

```
kabir = Student("Kabir", 4)
```

```
kabir.introduce_self()
```

- kabir is an Object of the Student Class
- It has attributes name="Kabir" and grade=4
- kabir can do actions like introduce_self()



Why Do We Need Classes?

- Organize our code like a tidy cupboard
 - Make many objects easily from one blueprint
- Keep code neat and easy to understand





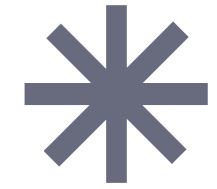
The Blueprint Challenge Quiz



Let's see what you remember with the
Blueprint Challenge!

- Answer questions about classes, objects, attributes, and methods
- Use your imagination and Indian examples





Blueprint Challenge Quiz Questions



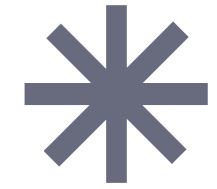
1. Which is the Class: Cake recipe or Chocolate cake?
2. Name 3 attributes of a School Bus object.
3. What does a Remote Control do? Give one example method.
4. True or False: An Object is a blueprint.



Summary: What We Learned Today

Classes are blueprints — like cookie cutters or recipes. Objects are real things made using blueprints. Objects have attributes (what they have) and methods (what they do). We coded a Student class and created Kabir!





Thank You & See You Next Time!

Thank you for learning with us! Keep
imagining and creating your own blueprints.
Planrs Academy

